Probability and Statistics

Joint Probability Distributions and Random Samples

(PART 2)

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1 The Distribution of the Sample Mean

2 The Distribution of a Linear Combination

The Distribution of the Sample Mean

- The distribution of the sample mean

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with mean value μ and standard deviation σ. Then

1. $E(\bar{X}) = \mu_{\bar{X}} = \mu$
2. $V(\bar{X}) = \sigma_{\bar{X}}^2 = \sigma^2/n$ and $\sigma_{\bar{X}} = \sigma/\sqrt{n}$

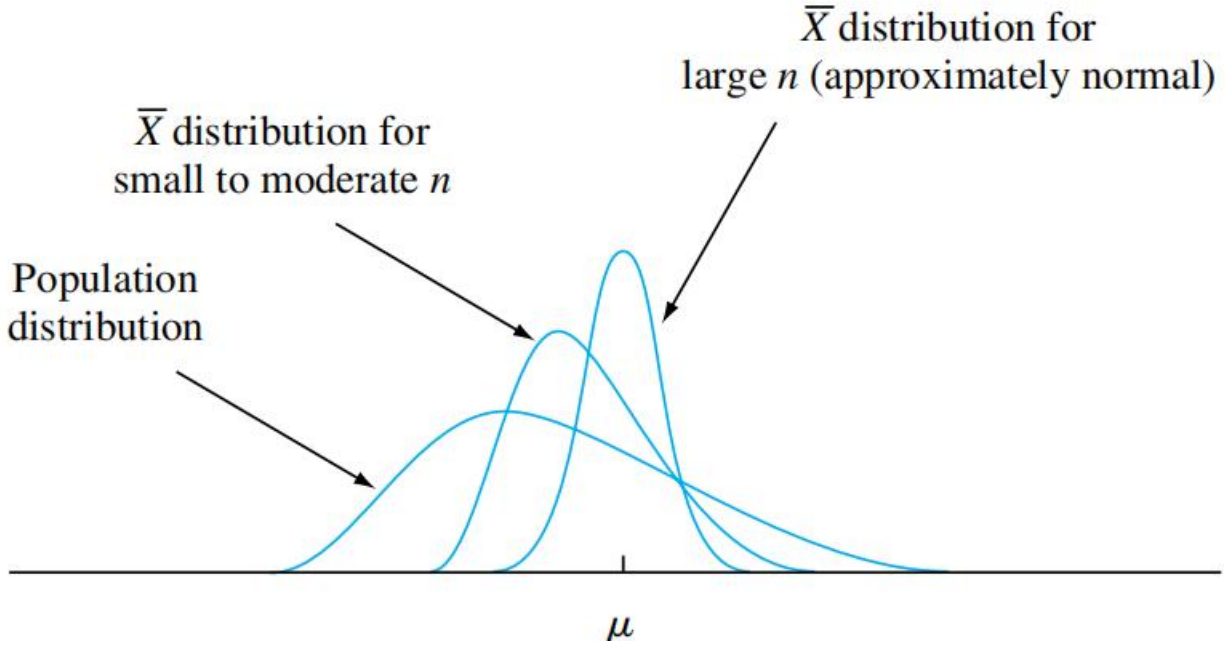
In addition, with $T_o = X_1 + \cdots + X_n$ (the sample total), $E(T_o) = n\mu$, $V(T_o) = n\sigma^2$, and $\sigma_{T_o} = \sqrt{n}\sigma$.



The Distribution of the Sample Mean

- Central limit theorem (中心极限定理)

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with mean μ and variance σ^2. Then if n is sufficiently large, $\overline{X}$ has approximately a normal distribution with $\mu_{\overline{X}} = \mu$ and $\sigma^2_{\overline{X}} = \sigma^2/n$, and T_o also has approximately a normal distribution with $\mu_{T_o} = n\mu$, $\sigma^2_{T_o} = n\sigma^2$. The larger the value of n, the better the approximation.



The Distribution of the Sample Mean

- Central limit theorem (中心极限定理)
- Assume that the random variable S_n is the sum of 48 independent experimental values of the random variable X whose PDF is given below, find the probability that S_n lies in the range $108 \leq S_n \leq 126$

$$f_X(x) = \begin{cases} \frac{1}{3} & 1 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$



The Distribution of the Sample Mean

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The expected value and variance of X are given by

$$E[X] = \frac{(4+1)}{2} = 2.5$$

$$\sigma_X^2 = \frac{(4-1)^2}{12} = \frac{3}{4}$$

Thus, the mean and variance of S_n are given by

$$E[S_n] = 48E[X] = (48)(2.5) = 120$$

$$\sigma_{S_n}^2 = 48\sigma_X^2 = (48)\left(\frac{3}{4}\right) = 36$$

Assuming that the sum approximates the normal random variable, which is usually true for $n \geq 30$, the CDF of the normalized random value of S_n becomes

$$P[S_n \leq s] = F_{S_n}(s) = \Phi\left(\frac{s - E[S_n]}{\sigma_{S_n}}\right) = \Phi\left(\frac{s-120}{6}\right)$$

Therefore, the probability that S_n lies in the range $108 \leq S_n \leq 126$ is given by

$$\begin{aligned} P[108 \leq S_n \leq 126] &= F_{S_n}(126) - F_{S_n}(108) \\ &= \Phi\left(\frac{126-120}{6}\right) - \Phi\left(\frac{108-120}{6}\right) \\ &= \Phi(1) - \Phi(-2) = \Phi(1) - \{1 - \Phi(2)\} \\ &= \Phi(1) + \Phi(2) - 1 \\ &= 0.8413 + 0.9772 - 1 \\ &= 0.8185 \end{aligned}$$

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The Distribution of the Sample Mean

- Central limit theorem (中心极限定理)

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution for which only positive values are possible $[P(X_i > 0) = 1]$. Then if n is sufficiently large, the product $Y = X_1 X_2 \cdot \cdots \cdot X_n$ has approximately a lognormal distribution.



1 The Distribution of the Sample Mean

2 The Distribution of a Linear Combination

The Distribution of the Sample Mean

- Linear combination of random variables

Given a collection of n random variables $X_1, \ldots, X_n$ and n numerical constants $a_1, \ldots, a_n$, the rv

$$Y = a_1X_1 + \cdots + a_nX_n = \sum_{i=1}^{n} a_iX_i$$

is called a **linear combination** of the X_is.

The Distribution of the Sample Mean

- Linear combination of random variables

Let $X_1, X_2, \ldots, X_n$ have mean values $\mu_1, \ldots, \mu_n$, respectively, and variances $\sigma_1^2, \ldots, \sigma_n^2$, respectively.

1. Whether or not the X_is are independent,

$$\begin{aligned} E(a_1X_1 + a_2X_2 + \cdots + a_nX_n) &= a_1E(X_1) + a_2E(X_2) + \cdots + a_nE(X_n) \\ &= a_1\mu_1 + \cdots + a_n\mu_n \end{aligned}$$

2. If $X_1, \ldots, X_n$ are independent,

$$\begin{aligned} V(a_1X_1 + a_2X_2 + \cdots + a_nX_n) &= a_1^2V(X_1) + a_2^2V(X_2) + \cdots + a_n^2V(X_n) \\ &= a_1^2\sigma_1^2 + \cdots + a_n^2\sigma_n^2 \end{aligned}$$

and

$$\sigma_{a_1X_1+\cdots+a_nX_n} = \sqrt{a_1^2\sigma_1^2 + \cdots + a_n^2\sigma_n^2}$$

3. For any $X_1, \ldots, X_n$,

$$V(a_1X_1 + \cdots + a_nX_n) = \sum_{i=1}^{n}\sum_{j=1}^{n} a_ia_j\,\mathrm{Cov}(X_i, X_j)$$



The Distribution of the Sample Mean

- Difference between two random variables

$E(X_1 - X_2) = E(X_1) - E(X_2)$ and, if X_1 and X_2 are independent, $V(X_1 - X_2) = V(X_1) + V(X_2)$.



Home work

- Ex. 9, Ex.19, Ex.38, Ex.48